Nonparametric combinatorial sequence models

Fabian L. Wauthier, UC Berkeley

with Nebojsa Jojic (MSR) and Michael I. Jordan (UCB)

30$^{\mathrm{th}}$ March, 2011

Biological motivation: Sequence variability

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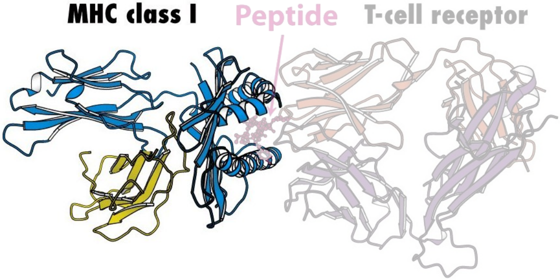
- Partial, long-range site dependencies

Example: MHC I proteins



Freeman and Company, 2007

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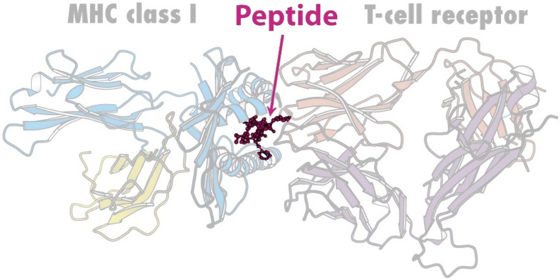

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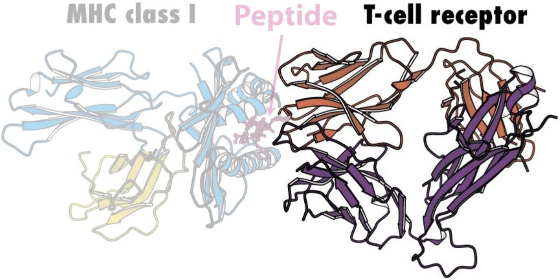

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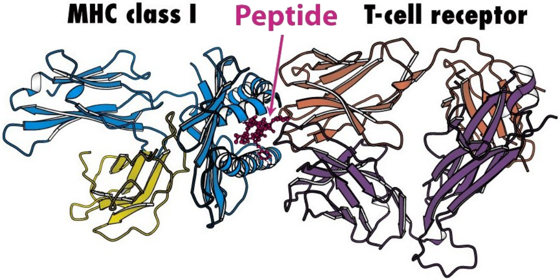

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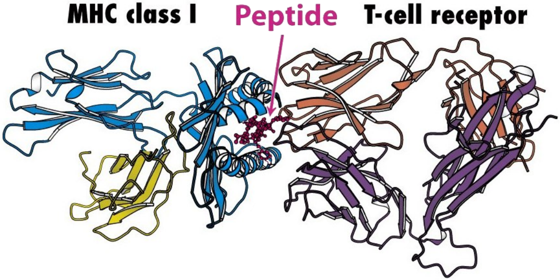

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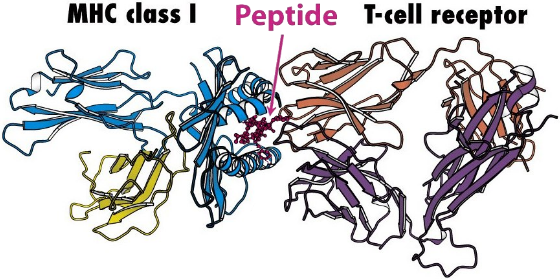

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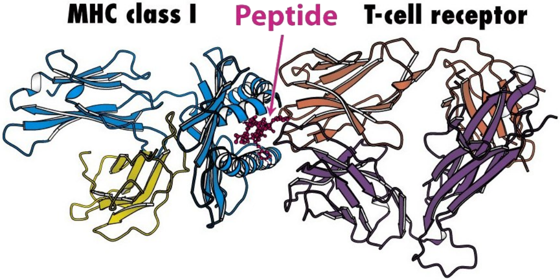

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Our Interest: model sequence **variability**, not its **origins**.

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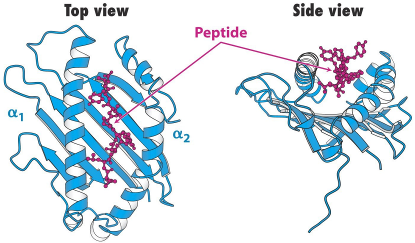

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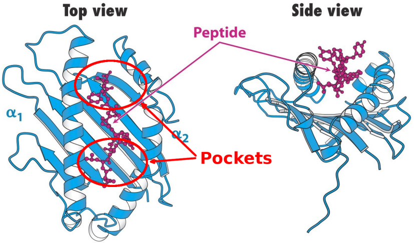

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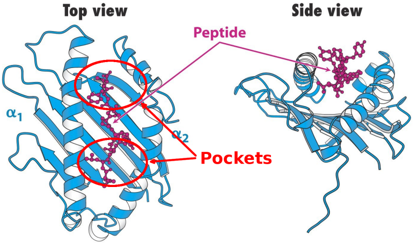

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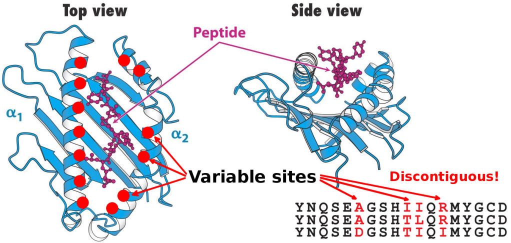

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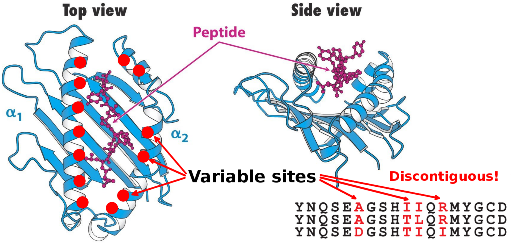

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  $\Longrightarrow$ **Markovian analysis inappropriate**

Our model: high level

A	T	G	C	G
C	T	G	T	G
A	C	A	C	G
A	C	A	C	A

Main idea: Each sequence is composed of smaller components.

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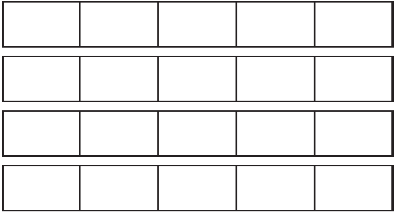

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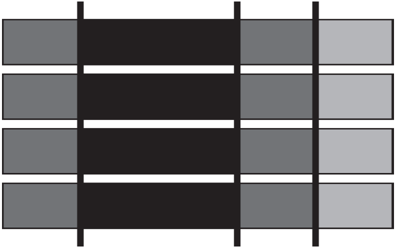

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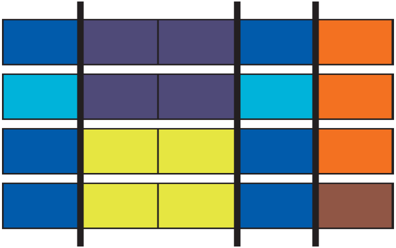

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C.f. Probabilistic index map (Jojic and Caspi, CVPR 2004; Jojic et al., UAI 2004)

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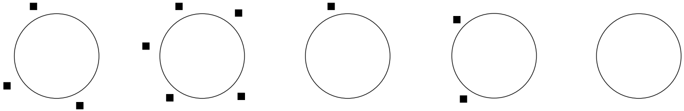
- **Our model**: A Chinese Restaurant Franchise (CRF) conditioned on a Chinese Restaurant Process (CRP)
  1. CRP: induces prior on number of site groups.
  2. CRF: induces prior on number of PSSMs and shares them among sequences.

Background: The Chinese Restaurant Process (CRP)



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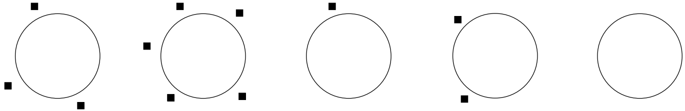

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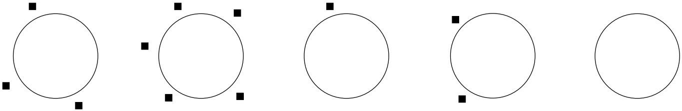

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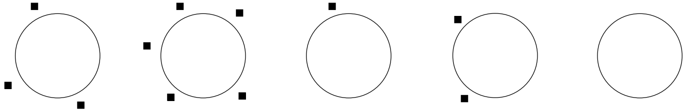

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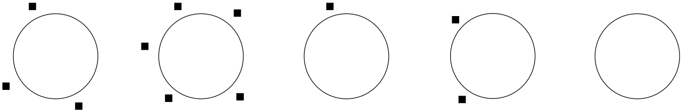

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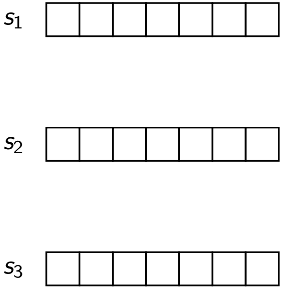

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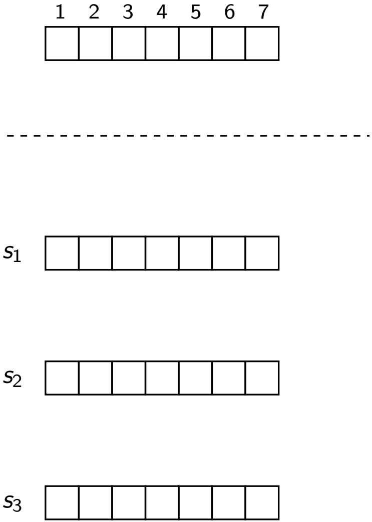

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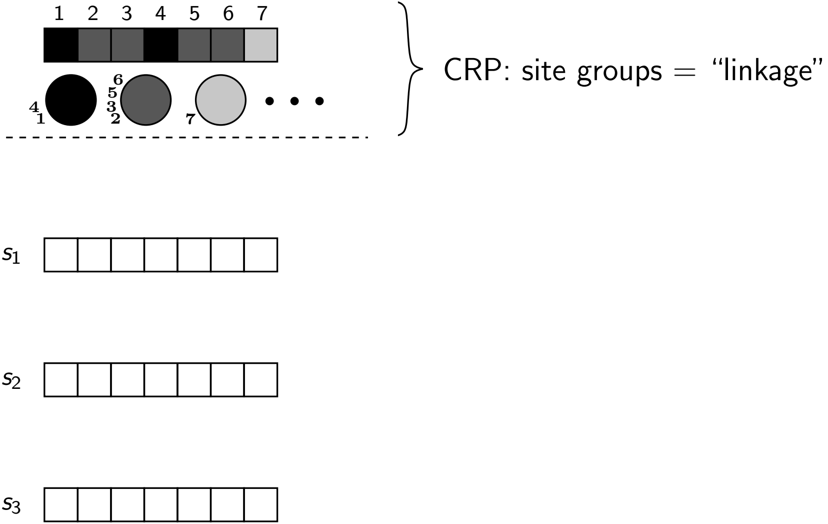

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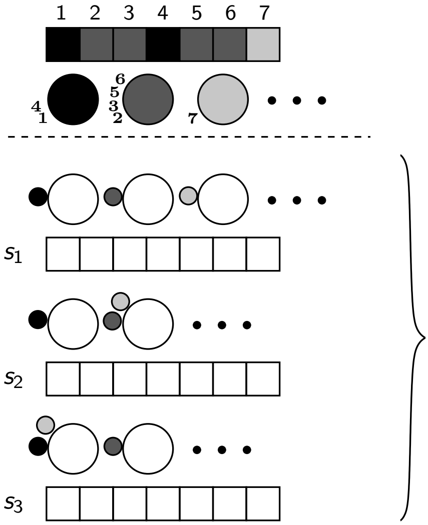

CRF: secondary site grouping

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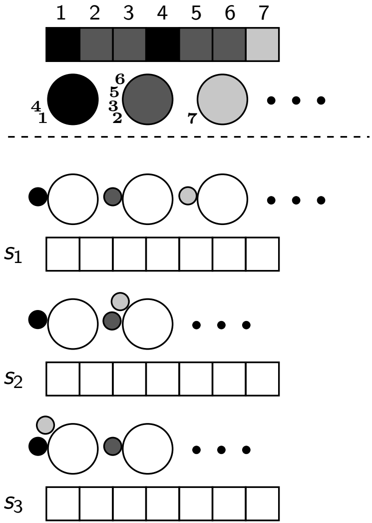

Parameters

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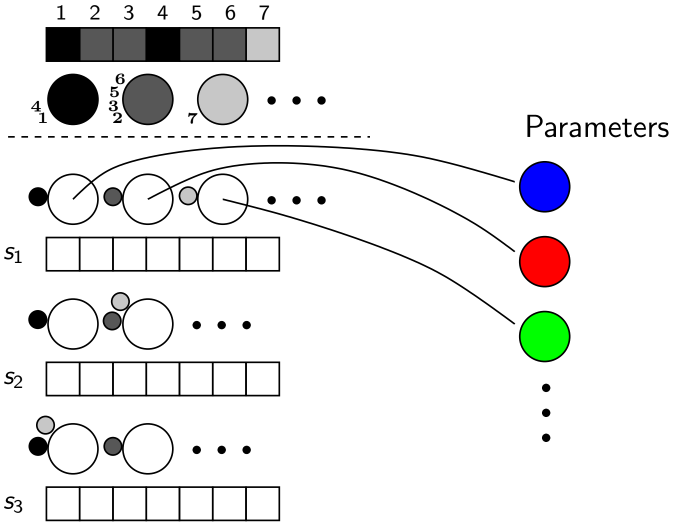

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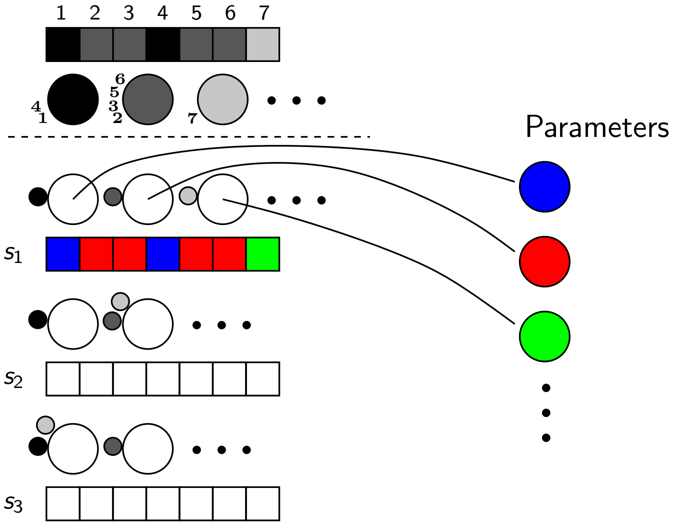

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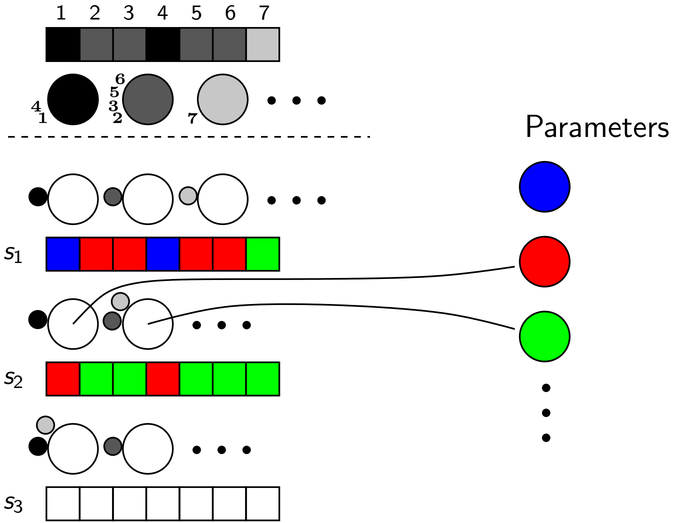

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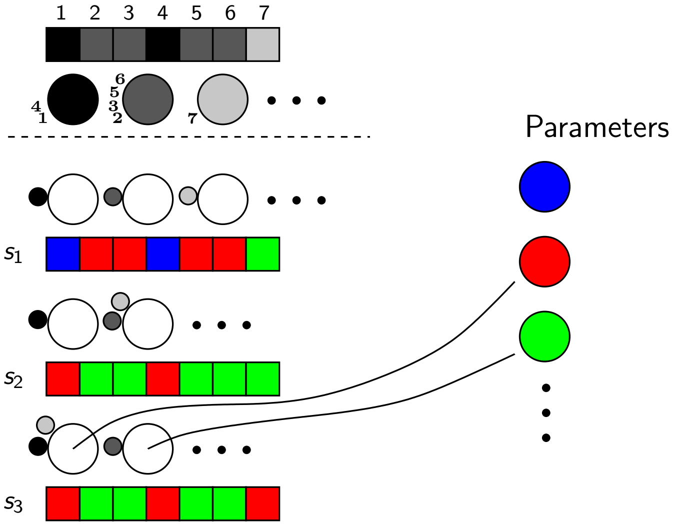

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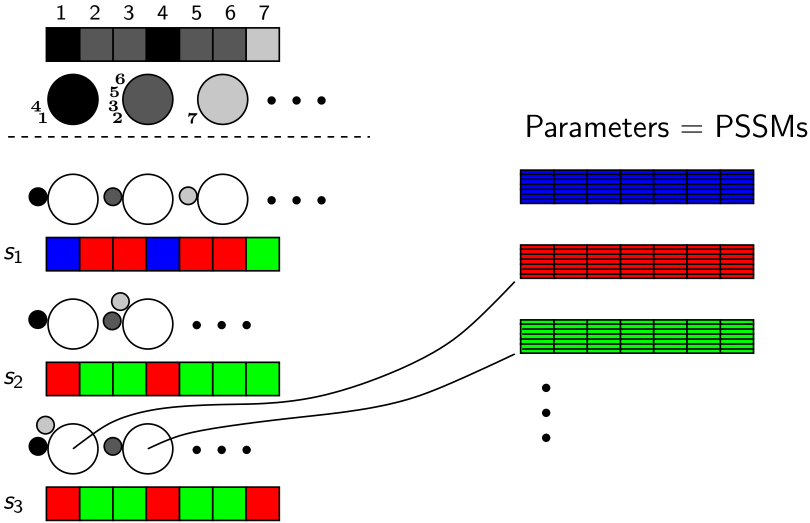

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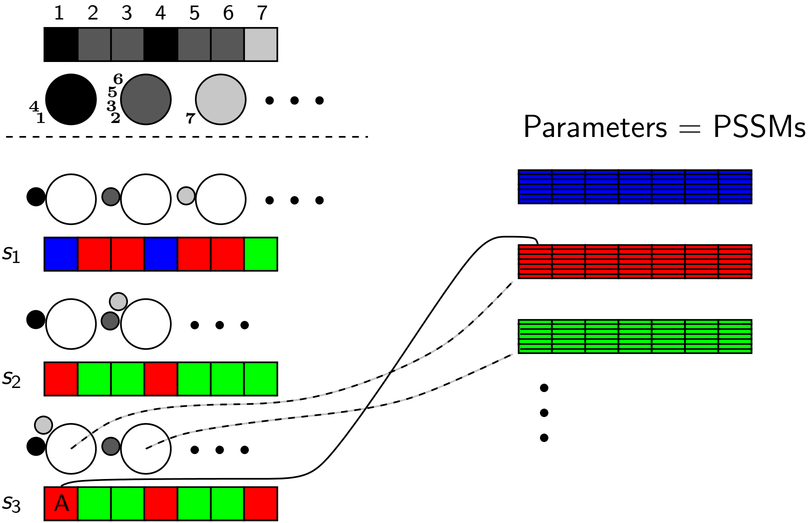

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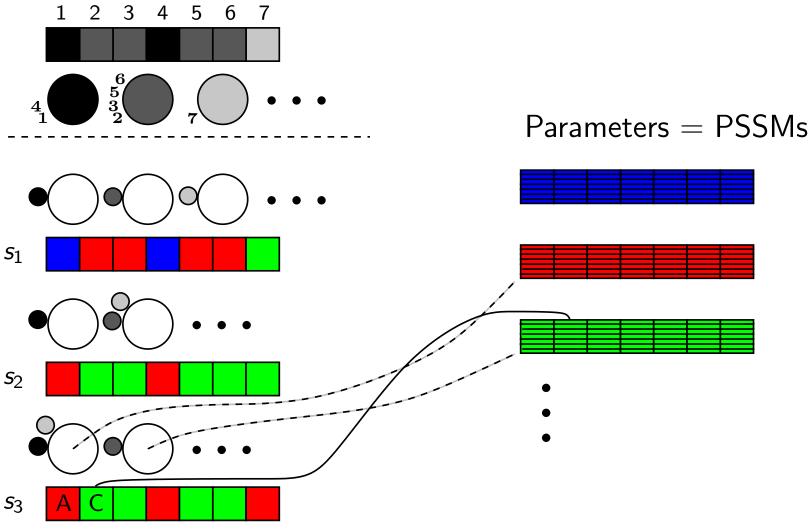

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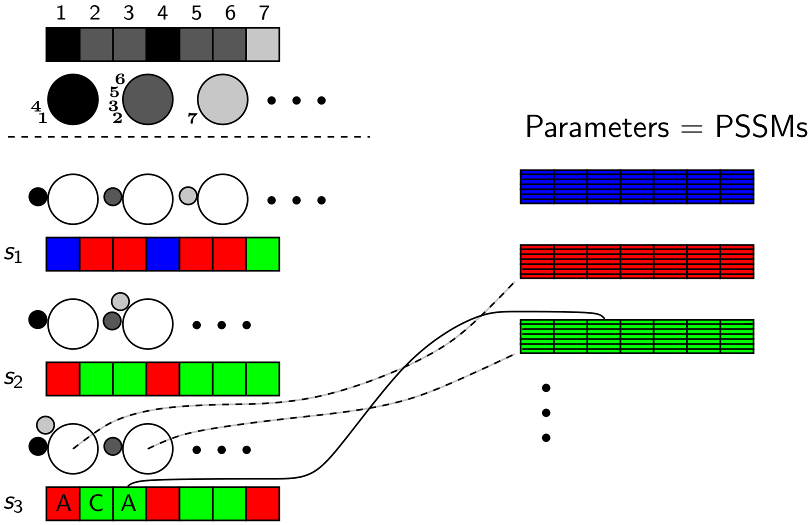

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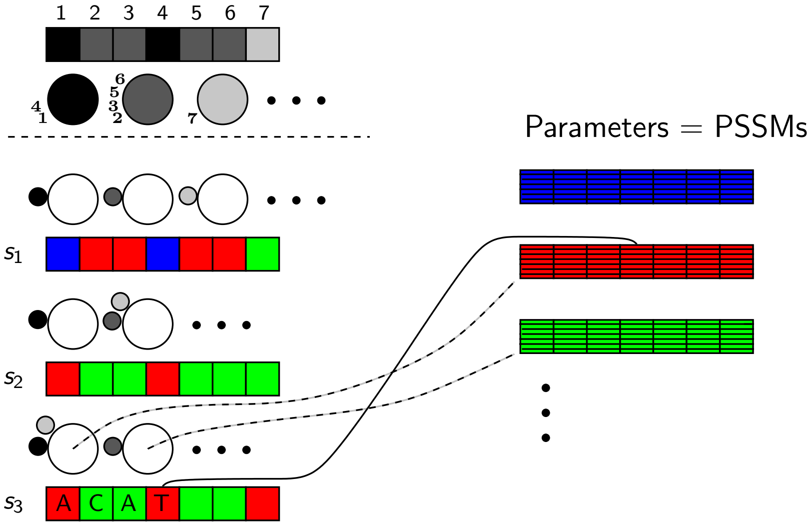

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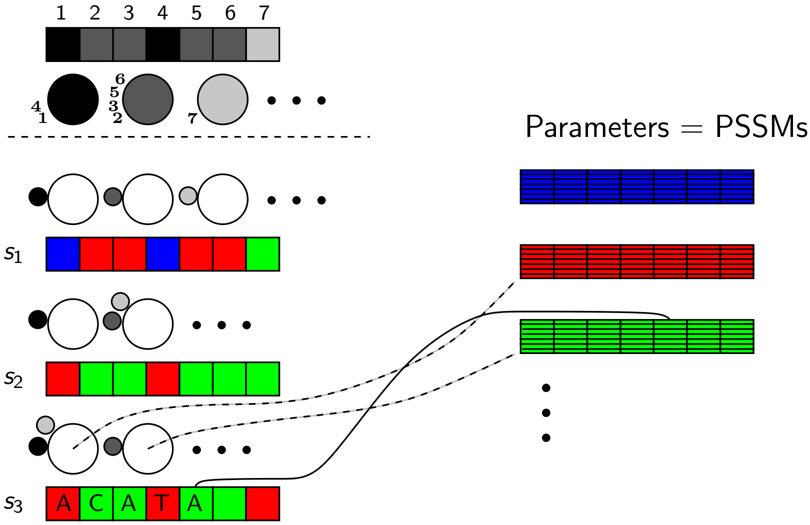

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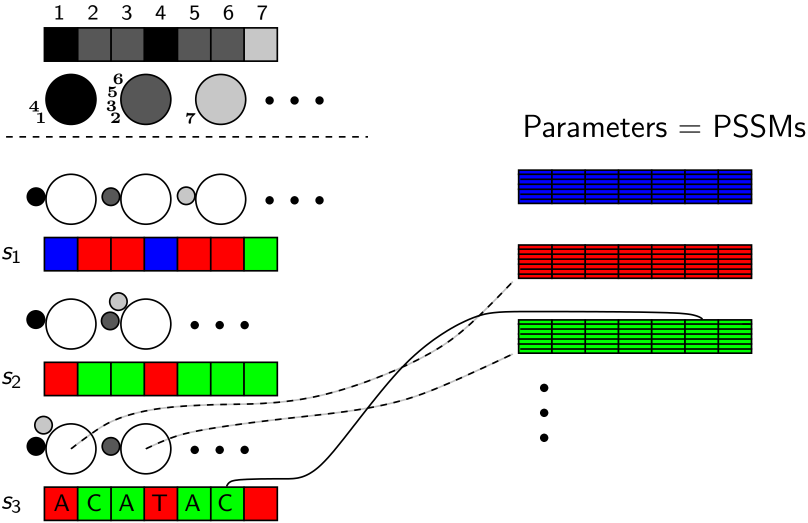

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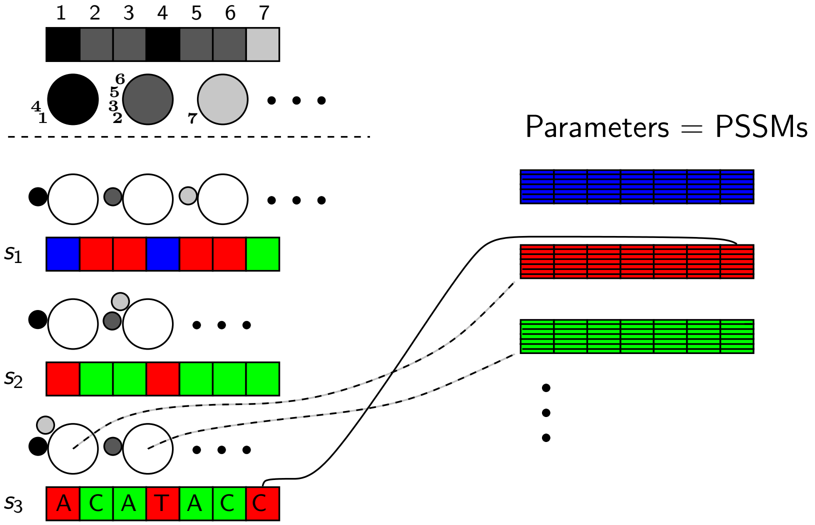

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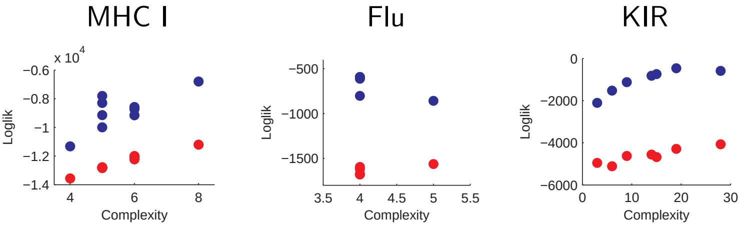
- Look at three datasets: blue = our model, red = mixture.



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  - Learn Θ_k for each sample k. Then average predictions.
  - Predict binding/non-binding peptides $\Rightarrow$ AUC score

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Transfer only	0.7285
Random	0.5

- Compare with state of the art (SOA) (Peters et al., 2006):

Method	AUC
MAP	0.8378
Averaging	0.8911
SOA	0.85–0.91

- Similar performance, but use only limited information:
  no spatial proximity, chemical properties, interaction features, nonlinearities.

Questions?